

Deep Learning-Based Early Stroke Detection Using CT Images

A Project Progress Report

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Certificate of Completion

This is to certify that the work entitled, “**Deep Learning-Based Early Stroke Detection Using CT Images**” is the bonafide work of **M.Ajay Srinivas(ID : N210142)**, **K.Sushanth(ID : N210274)**, **Y.Shanmukh Satya Sai(ID : N210095)** and **D.Revanth(ID : N211086)** carried out under my guidance and supervision for the 3rd-year project of Bachelor of Technology in the department of Computer Science and Engineering under RGUKT IIIT Nuzvid. This work is done during the January 2026 - April 2026 academic session under our guidance.

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DECLARATION

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We also declare that this project is a result of our own effort and has not been copied or imitated from any source. Citations from any websites are mentioned in the references. The results embodied in this project report have not been submitted to any other university or institute for the award of any degree or diploma.

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ABSTRACT

Stroke is one of the leading causes of death and long-term disability worldwide, where early and accurate detection plays a crucial role in improving patient outcomes. This project presents a deep learning-based approach for early stroke detection using CT brain images. The proposed system integrates advanced feature extraction techniques with both machine learning and deep learning models to enhance diagnostic performance. In this study, features are extracted from CT images using pre-trained convolutional neural network architecture, InceptionV3 . These features are then used to train and evaluate multiple classification models, including Logistic Regression, Support Vector Machine (SVM), Random Forest, XGBoost, Long Short-Term Memory (LSTM), and Bidirectional LSTM (BiLSTM). A comprehensive comparative analysis is performed to assess the performance of each model based on evaluation metrics such as accuracy, precision, recall, and F1-score. The results demonstrate that deep learning models, particularly BiLSTM, outperform traditional machine learning approaches. Additionally, feature extraction using InceptionV3 provides better performance compared to other options. The proposed approach improves the accuracy and efficiency of stroke detection, reduces dependency on manual diagnosis, and supports faster clinical decision-making. This system can assist healthcare professionals in early diagnosis and has the potential to be integrated into real-time medical applications for improved patient care.

Contents

1	Introduction	8
1.1	Background of the Study	8
1.2	Motivation	8
1.3	Objectives	8
1.4	Scope of the Project	9
2	Literature Survey	10
2.1	Related Works	10
2.2	Introduction	15
2.3	Research Gaps	16
3	Proposed System	17
3.1	Problem Definition	17
3.2	Framework of Proposed Model	19
3.3	Methodology	21
3.3.1	Data Preprocessing	21
3.3.2	Feature Extraction	22
3.3.3	Feature Optimization	22
3.3.4	Model Training	23
3.3.5	Model Evaluation	24
3.3.6	Final Prediction	24
4	Results & Comparative Analysis	25
4.1	Data Collection	25
4.2	Results obtained using ML Models	26
4.3	Results Obtained Using DL Models	28
4.4	Error Analysis	29
5	Conclusion and Future Work	36

List of Figures

1	Flow of the Project	19
2	Images in the Dataset	25
3	Logistic Regression Confusion Matrix	29
4	Logistic Regression ROC Curve	29
5	XGBoost Confusion Matrix	30
6	XGBoost ROC Curve	30
7	Random Forest Confusion Matrix	31
8	Random Forest ROC Curve	31
9	SVM Confusion Matrix	32
10	SVM ROC Curve	32
11	LSTM Confusion Matrix	33
12	LSTM ROC Curve	33
13	BiLSTM Confusion Matrix	34
14	BiLSTM ROC Curve	34

1 Introduction

1.1 Background of the Study

Stroke is one of the leading causes of death and long-term disability worldwide. It occurs when the blood supply to the brain is interrupted or reduced, preventing brain tissue from receiving oxygen and nutrients. This can lead to severe brain damage or even death if not treated promptly. Early detection of stroke is therefore critical in improving patient survival rates and reducing long-term complications.

Computed Tomography (CT) imaging is widely used in medical practice for the diagnosis of stroke due to its speed, availability, and effectiveness in identifying abnormalities in the brain. However, traditional diagnosis relies heavily on the expertise of radiologists, making the process time-consuming and prone to human error.

With the advancement of artificial intelligence and deep learning techniques, there is a growing interest in developing automated systems for medical image analysis. These systems can assist healthcare professionals by providing faster and more accurate detection of stroke, ultimately improving clinical outcomes and supporting timely decision-making.

1.2 Motivation

Stroke is a critical medical condition that requires immediate diagnosis and treatment to prevent severe brain damage and reduce mortality rates. However, in many cases, early detection is delayed due to the time-consuming nature of manual analysis and the limited availability of experienced radiologists, especially in rural and underdeveloped areas.

Traditional diagnostic methods rely heavily on human expertise, which can lead to variability in interpretation and potential errors. Even small delays in identifying stroke can significantly impact treatment effectiveness and patient recovery. Therefore, there is a strong need for a system that can assist in faster and more accurate diagnosis.

Recent advancements in artificial intelligence and deep learning have shown great potential in medical image analysis. These technologies can automatically extract meaningful patterns from complex data and improve diagnostic accuracy. This motivates the development of an automated stroke detection system that can support healthcare professionals, reduce workload, and enhance decision-making.

1.3 Objectives

The primary objective of this project is to develop an efficient and accurate system for early stroke detection using CT brain images. The system aims to leverage advanced deep learning techniques to assist in timely diagnosis and improve patient outcomes.

The specific objectives of the project are as follows:

- To develop an automated system for detecting stroke from CT scan images.
- To extract meaningful features from medical images using deep learning models such as InceptionV3 and VGG19.
- To apply a Genetic Algorithm (GA) for selecting the most relevant features and reducing redundancy.
- To implement Bidirectional Long Short-Term Memory (BiLSTM) for accurate classification of stroke and normal cases.
- To perform a comparative analysis of multiple machine learning and deep learning models.
- To evaluate the performance of the system using metrics such as accuracy, precision, recall, and F1-score.

1.4 Scope of the Project

The scope of this project is to develop an automated system for early detection of brain stroke using CT scan images. The system focuses on applying deep learning and machine learning techniques to assist in accurate and timely diagnosis.

The project primarily deals with binary classification, where CT brain images are classified into two categories: Normal and Stroke. Pre-trained deep learning models such as VGG19 and InceptionV3 are used for feature extraction, followed by a Genetic Algorithm for selecting the most relevant features. These optimized features are then used to train multiple classification models, including Logistic Regression, Support Vector Machine, Random Forest, XGBoost, LSTM, and BiLSTM.

The system is designed to:

- Improve the accuracy of stroke detection
- Reduce manual effort in analyzing CT images
- Assist medical professionals in decision-making

This project is limited to:

- CT scan images only (not MRI or other imaging modalities)
- A specific dataset
- Offline analysis (not real-time clinical deployment)

Although the system provides promising results, it is intended for research and educational purposes and not for direct clinical use without further validation.

Future extensions may include real-time deployment, integration with hospital systems, and expansion to multi-class classification of different stroke types

2 Literature Survey

2.1 Related Works

Chikmurge et al. proposed that early detection of brain stroke using deep learning techniques is critical for improving diagnostic accuracy and reducing mortality rates.

Early detection of brain stroke is critical for reducing mortality rates and improving patient recovery outcomes. Stroke remains one of the leading causes of disability and death worldwide, making timely and accurate diagnosis essential for effective medical intervention. Traditional diagnostic approaches, which rely on manual examination of CT brain scans by radiologists, are often time-consuming and may be affected by subjective interpretation and human error.

To address these limitations, this study proposes an automated deep learning-based framework for early stroke detection using CT brain images. The proposed system integrates a Genetic Algorithm (GA) for optimal feature selection and parameter optimization with a Bidirectional Long Short-Term Memory (BiLSTM) network for accurate classification. The Genetic Algorithm enhances model performance by selecting the most discriminative features, reducing redundancy, and improving generalization capability. The BiLSTM model captures complex spatial and sequential patterns within CT image features, enabling more precise classification.

A binary classification dataset consisting of normal and stroke CT images was used to train and evaluate the proposed model. Experimental results demonstrate that the GA-optimized BiLSTM model achieves high classification accuracy, along with strong precision, recall, and F1-score values. The confusion matrix further indicates balanced performance across both classes, highlighting the robustness and reliability of the proposed approach.

The findings suggest that combining evolutionary optimization techniques with deep learning architectures can significantly enhance stroke detection accuracy. Future work will focus on expanding the dataset, improving model generalization, and integrating the system into real-time clinical environments to support radiologists in making faster and more accurate diagnostic decisions.

Thirumoorthy et al. proposed that convolutional neural networks can be effectively used for early detection of ischemic stroke, improving diagnostic accuracy.

Ischemic brain stroke is a major cause of mortality and long-term neurological disability worldwide, requiring rapid and precise diagnosis to minimize irreversible brain damage. Conventional stroke detection primarily depends on radiologists' interpretation of CT brain images, a process that is time-consuming and prone to subjective variability.

ity. Delays in diagnosis can significantly reduce the effectiveness of thrombolytic therapy, leading to severe neurological impairment or death.

To address these challenges, this study proposes an automated deep learning-based framework for early ischemic stroke detection using CT brain images. The system processes medical images in DICOM format and converts them into standardized formats for efficient analysis. Advanced image preprocessing techniques are applied to enhance image quality and extract meaningful features. A Genetic Algorithm (GA) is employed for optimal feature selection and parameter tuning, improving model efficiency and reducing redundancy. The selected features are then classified using a Bidirectional Long Short-Term Memory (BiLSTM) network, which effectively captures complex spatial dependencies within CT image representations.

The proposed model performs binary classification to distinguish between normal (benign) cases and acute ischemic stroke (malignant) cases. Experimental evaluation demonstrates high accuracy, along with strong precision, recall, and F1-score values, indicating reliable and balanced performance.

By integrating evolutionary optimization with deep learning, the proposed system enhances diagnostic speed, reduces radiologist workload, and supports timely clinical decision-making. This approach contributes to improved patient outcomes, more efficient treatment planning, and increased accessibility to accurate stroke diagnosis in real-world healthcare settings.

Venkadesh et al. developed a deep learning-based stroke detection system using CNN and federated learning, enhancing data privacy and model performance.

Brain stroke is one of the leading causes of mortality and long-term disability worldwide, where early and accurate detection is critical for effective clinical intervention. Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have significantly improved the performance of automated stroke detection from CT brain images. However, conventional centralized training approaches require aggregating sensitive patient data in a single location, raising serious privacy and security concerns.

To address these limitations, this study proposes a hybrid framework that integrates Convolutional Neural Networks (CNNs) with Federated Learning (FL) for privacy-preserving brain stroke detection and localization. The CNN model is employed to extract discriminative features from CT images and perform binary classification into stroke and non-stroke categories. Federated Learning enables collaborative model training across multiple medical institutions without sharing raw patient data, ensuring data confidentiality while benefiting from diverse and distributed datasets.

Experimental results demonstrate that the proposed Hybrid CNN-FL framework achieves an accuracy of 94%, outperforming traditional centralized CNN models (89%) and conventional machine learning approaches (84%). The integration of FL not only maintains high precision and recall but also enhances model robustness, scalability, and generalization across heterogeneous datasets.

The findings highlight that combining CNNs with Federated Learning provides a powerful and privacy-aware solution for large-scale stroke detection. This approach supports real-time, secure, and collaborative medical imaging applications while preserving sensitive healthcare data. Future work will focus on improving the communication efficiency of federated frameworks and extending the methodology to other critical medical imaging tasks.

Monika et al. conducted a comparative analysis of various deep learning models for brain stroke detection, highlighting performance differences across architectures.

Early-stage stroke prediction plays a critical role in minimizing neurological damage and improving patient recovery outcomes. Accurate classification of stroke in CT brain imaging remains a challenging task due to subtle brain abnormalities and variations in imaging characteristics. Therefore, rapid and reliable diagnostic systems are essential to support clinical decision-making and ensure timely intervention.

This study presents a comparative evaluation of the ERDP-CNN (Enhanced Region Detection and Processing Convolutional Neural Network) model for automated stroke detection using CT brain images. The proposed approach integrates advanced feature extraction and feature selection techniques to optimize dimensionality, reduce redundancy, and minimize classification errors. The performance of the ERDP-CNN model is compared with several traditional and contemporary machine learning methods, including Logistic Regression (LR), Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Decision Tree (DT), and OzNet. Experimental results demonstrate that the ERDP-CNN model significantly outperforms the existing methods across multiple evaluation metrics.

The model achieves an Accuracy of 98.9%, Precision of 97.03%, Sensitivity of 95.56%, and an F1-Score of 99.98%. These results indicate superior classification capability, improved robustness, and enhanced reliability. The implementation and validation were conducted using a MATLAB simulation environment, confirming the model's effectiveness for early stroke diagnosis.

The findings highlight the potential of the ERDP-CNN framework to enhance medical decision-making in CT-based stroke detection and suggest promising applicability to other medical imaging domains.

Li and Yang proposed a CNN-BiLSTM neural network model to improve prediction accuracy, demonstrating the effectiveness of combining spatial and sequential learning.

Load fluctuations in core central business district (CBD) areas, driven by abrupt meteorological variations and dynamic passenger flow, often lead to energy scheduling mismatches and operational inefficiencies. To address this issue, this paper proposes a short-term electric load forecasting model based on a hybrid Convolutional Neural Network–Bidirectional Long Short-Term Memory (CNN–BiLSTM) architecture. The raw load data are first preprocessed through data cleaning, missing value imputation, and normalization to ensure data integrity and stability. A CNN is then employed to extract high-level spatial correlation features, while a BiLSTM network captures bidirectional temporal dependencies and cross-cycle characteristics in the time series data.

The proposed model is evaluated against a standalone BiLSTM baseline through comparative experiments. Results demonstrate that the CNN–BiLSTM model reduces the Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) by 6.9% and 12.1%, respectively, and improves the coefficient of determination (R^2) by 0.2 percentage points. These findings confirm the effectiveness of the proposed hybrid framework in modeling complex spatiotemporal patterns and enhancing short-term load forecasting accuracy.

Saleem et al. introduced a machine learning-based diagnostic model using neuroimages, focusing on improving stroke identification through advanced feature analysis.

Cerebrovascular diseases, particularly stroke, are among the leading causes of mortality and long-term disability worldwide. Despite being preventable and treatable, delayed diagnosis significantly increases the risk of severe neurological damage and poor clinical outcomes.

Early detection and rapid intervention are therefore critical for reducing disease burden and improving patient recovery. In recent years, machine learning and deep learning techniques have gained considerable attention for their potential in automated stroke detection from medical imaging data.

This study proposes an early stroke detection system based on CT brain images integrated with a Genetic Algorithm (GA) and a Bidirectional Long Short-Term Memory (BiLSTM) network. The Genetic Algorithm is employed for optimal feature selection, identifying the most relevant and discriminative features to enhance classification performance while reducing dimensionality. These optimized features are then provided to the BiLSTM model, which effectively captures complex patterns within the extracted image representations.

The system’s performance was evaluated using cross-validation and multiple metrics, including Accuracy, Precision, Recall, F1-Score, Receiver Operating Characteristic (ROC) curve, and Area Under the Curve (AUC). The proposed model achieved an accuracy of 96.5%, demonstrating strong classification capability and reliability. Furthermore, comparative analysis with traditional machine learning models such as Logistic Regression, Decision Trees, Random Forests, Naïve Bayes, and Support Vector Machines confirmed the superior performance of the proposed GA-BiLSTM framework.

The results indicate that the proposed diagnostic system can effectively support physicians in making informed and timely decisions for stroke detection and treatment planning.

Pang et al. proposed the concept of convolution within convolution to enhance feature extraction capability in deep neural networks.

Network in Network (NiN) is a significant extension of deep convolutional neural networks (CNNs), composed of alternating convolutional and pooling layers. Unlike conventional CNNs that rely on linear convolutional filters, NiN replaces them with shallow multilayer perceptrons (MLPs), introducing nonlinearity and enhancing feature representation capability. The integration of MLPs and 1×1 convolutions strengthens spatial modeling and improves recognition performance. However, the dense fully connected structure within shallow MLPs substantially increases the number of parameters, leading to higher computational complexity and potential overfitting.

To address this limitation, we propose a novel architecture termed Convolution in Convolution (CiC), which replaces dense shallow MLPs with sparse shallow MLPs. In the proposed approach, one or more layers within the shallow MLP are sparsely connected either along the channel dimension or across the combined channel-spatial domain. This is achieved by applying unshared convolution across the channel dimension while maintaining shared convolution across the spatial dimension in selected computational layers. The proposed sparsification strategy significantly reduces parameter redundancy while preserving expressive power.

Experimental evaluations conducted on the CIFAR-10, augmented CIFAR-10, and CIFAR-100 datasets demonstrate that the CiC architecture achieves competitive or improved classification performance with reduced parameter complexity, validating the effectiveness of the proposed method.

Zhong et al. introduced DualConv, which uses dual convolutional kernels to improve efficiency and performance in lightweight neural networks.

Convolutional neural networks (CNNs) typically demand substantial memory and computational resources, limiting their deployment on embedded and resource-constrained devices. To address this challenge, we propose Dual Convolutional Kernels (DualConv), a lightweight architectural module designed to reduce model complexity while maintaining or improving performance. DualConv simultaneously applies convolutional kernels to the same input feature map and leverages group convolution to efficiently organize filters, thereby enhancing representational capacity with reduced computational overhead. The proposed module is model-agnostic and can be integrated into existing CNN architectures, including VGG-16 and ResNet-50 for image classification, YOLO and R-CNN for object detection, and fully convolutional networks (FCNs) for semantic segmentation. In this study, we primarily evaluate DualConv on classification tasks, as these architectures serve as foundational backbones for many vision applications. Additionally, we assess its effectiveness in object detection using YOLOv3.

Extensive experiments demonstrate that DualConv substantially reduces parameter count and computational cost while achieving comparable or improved accuracy relative to baseline models. Specifically, integrating DualConv into MobileNetV2 reduces the number of parameters by 54% with only a 0.68% decrease in accuracy on the CIFAR-100 dataset. When parameter efficiency is not the primary constraint, DualConv improves MobileNetV1 accuracy by 4.11% on the same dataset. Furthermore, incorporating DualConv into YOLOv3 enhances detection speed and increases accuracy by 4.4% on the PASCAL VOC dataset. These results validate DualConv as an effective and flexible module for designing efficient deep neural networks suitable for real-world deployment in constrained environments.

2.2 Introduction

The literature survey highlights that significant research has been carried out in the field of stroke detection using machine learning and deep learning techniques. Various approaches such as Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), and hybrid models have been proposed to improve detection accuracy. However, many of these methods face challenges such as high computational complexity, lack of effective feature selection, and limited generalization across different datasets.

To address these limitations, there is a need for a more efficient and accurate approach that can handle high-dimensional medical image data while maintaining reliability and speed. The integration of feature extraction, optimization, and advanced classification techniques can significantly enhance the performance of stroke detection systems.

In this context, the proposed system combines Convolutional Neural Networks (CNN) for feature extraction, a Genetic Algorithm (GA) for selecting the most relevant features, and a Bidirectional Long Short-Term Memory (BiLSTM) network for classification. This integrated approach aims to improve accuracy, reduce redundancy, and support faster clinical decision-making.

2.3 Research Gaps

Although numerous studies have explored stroke detection using machine learning and deep learning techniques, several research gaps still exist.

Many existing works focus primarily on improving accuracy using individual models such as Logistic Regression, Support Vector Machine (SVM), Random Forest, or Convolutional Neural Networks (CNNs). However, limited attention has been given to integrating effective feature extraction, feature optimization, and advanced sequential classifiers into a single unified framework.

Most traditional machine learning approaches rely on handcrafted features, which may not fully capture complex spatial patterns in CT images. On the other hand, deep learning models such as VGG19 and InceptionV3 provide strong feature extraction capabilities but are often used without proper feature selection, leading to redundancy and increased computational complexity.

Additionally, while Long Short-Term Memory (LSTM) networks have been applied in some studies, the potential of Bidirectional Long Short-Term Memory (BiLSTM) for capturing contextual relationships in medical image features remains underexplored in stroke detection tasks.

Furthermore, many existing studies lack comprehensive comparative analysis across multiple machine learning and deep learning models using the same dataset, making it difficult to determine the most effective approach.

Therefore, there is a need for a hybrid, optimized, and comparative framework that integrates deep feature extraction, feature selection using Genetic Algorithm, and advanced classification using BiLSTM to improve performance, efficiency, and reliability in stroke detection.

3 Proposed System

3.1 Problem Definition

Let the dataset be defined as:

$$\mathbf{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N \quad (3.1)$$

where $\mathbf{x}_i$ represents the i^{th} CT brain image and $y_i \in \{0, 1\}$ is the corresponding class label, with 0 denoting Normal and 1 denoting Stroke.

Objective

The objective is to learn a mapping function:

$$f : \mathcal{X} \rightarrow \mathcal{Y} \quad (3.2)$$

such that:

$$\hat{y}_i = f(\mathbf{x}_i) \quad (3.3)$$

where $\hat{y}_i$ is the predicted class label.

Optimization Goal

The model aims to minimize the classification loss:

$$\min \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{y}_i) \quad (3.4)$$

where $L(\cdot)$ denotes the loss function.

Feature Transformation

The input images are transformed into deep feature vectors using InceptionV3:

$$\mathbf{z}_i = \phi(\mathbf{x}_i) \quad (3.5)$$

where $\phi(\cdot)$ denotes the feature extraction function.

Dimensionality Reduction

Principal Component Analysis is applied as:

$$\mathbf{z}'_i = \text{PCA}(\mathbf{z}_i) \quad (3.6)$$

Feature Selection

Genetic Algorithm selects the optimal feature subset:

$$\mathbf{z}_i^* = \text{GA}(\mathbf{z}_i') \quad (3.7)$$

Final Prediction

The optimized features are used for classification:

$$\hat{y}_i = f(\mathbf{z}_i^*) \quad (3.8)$$

The objective is to learn an optimal mapping from CT images to class labels by minimizing classification error using extracted and optimized features.

3.2 Framework of Proposed Model

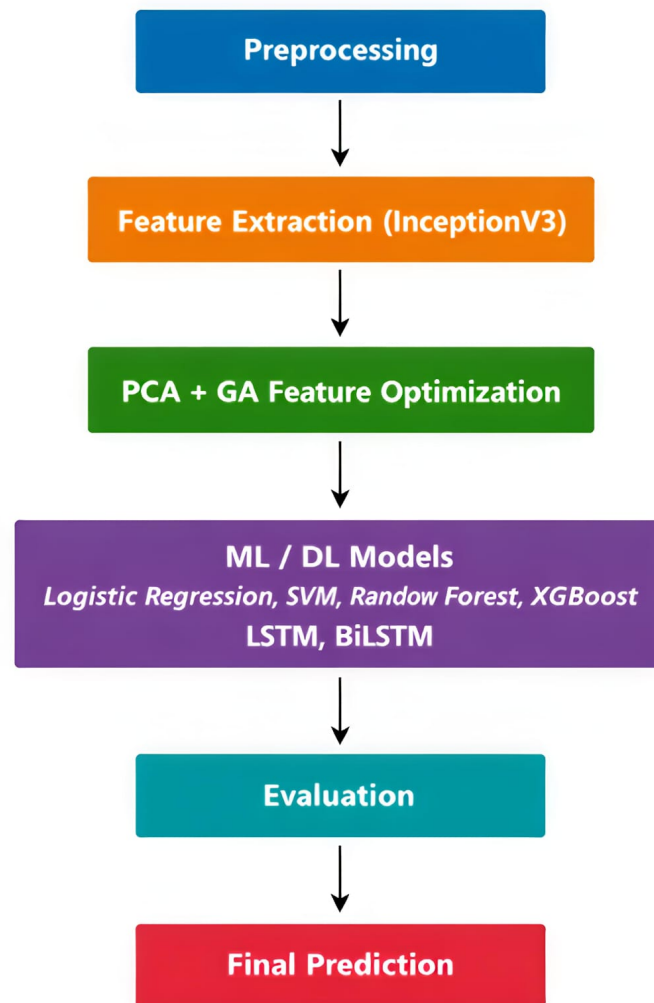


Figure 1: **Flow of the Project**

The proposed system follows a structured pipeline for efficient and accurate stroke detection using CT brain images. The framework is divided into multiple stages, each contributing to improving performance and reliability.

1. **Input: Brain CT Images**

The system takes CT brain images as input, consisting of two classes: normal and stroke. These images form the foundation for model training and evaluation.

2. **Data Splitting (Training and Testing)**

The dataset is divided into two parts:

- (a) Training Set → Used to train the model
- (b) Testing Set → Used to evaluate performance

3. Image Preprocessing

The input images are preprocessed to ensure uniformity and quality. This includes:

- (a) Resizing images to a fixed dimension (224×224)
- (b) Normalizing pixel values to a standard range
- (c) Preparing data for efficient model input

This step ensures compatibility with deep learning models and improves training stability.

4. Feature Extraction using CNN (InceptionV3)

Feature extraction is performed using InceptionV3, a pre-trained convolutional neural network. The model extracts rich and hierarchical features from the images, converting them into high-dimensional feature vectors (2048 features).

5. Feature Reduction and Selection

To optimize the feature space:

- Principal Component Analysis (PCA) is used to reduce dimensionality while preserving important information
- Genetic Algorithm (GA) is applied to select the most relevant features based on a fitness function features

This step reduces redundancy and improves model efficiency.

6. Model Training

The optimized feature set is passed to a Bidirectional Long Short-Term Memory (BiLSTM) model.

Processes data in:

- Forward Direction
- Backward Direction

BiLSTM is especially effective in handling complex patterns in medical data.

7. Evaluation

The model is evaluated using performance metrics such as:

- Accuracy
- Precision
- Recall
- F1-score

These metrics help assess the effectiveness of each model.

8. Output: Classification Results

The final stage produces the classification output, whether there is a Stroke or not.

"The proposed framework ensures a well-structured pipeline that combines deep feature extraction, optimization techniques, and advanced models to achieve accurate and efficient stroke detection."

3.3 Methodology

The proposed methodology follows a systematic pipeline for stroke detection using CT brain images. The framework integrates preprocessing, feature extraction, feature optimization, and classification to achieve accurate predictions.

3.3.1 Data Preprocessing

Image Resizing and Representation

All input CT images are resized to a fixed dimension of 224×224 pixels to ensure compatibility with the InceptionV3 model.

1. Image Resizing Mapping

$$I'(x', y') = I\left(\frac{x'}{s_x}, \frac{y'}{s_y}\right) \quad (3.9)$$

2. Scaling Factors

$$s_x = \frac{W'}{W}, \quad s_y = \frac{H'}{H} \quad (3.10)$$

3. Bilinear Interpolation

$$I'(x', y') = \sum_{i=0}^1 \sum_{j=0}^1 w_{ij} I(x_i, y_j) \quad (3.11)$$

4. Image Normalization

$$x' = \frac{x}{255} \quad (3.12)$$

5. General Image Representation

$$I \in R^{H \times W \times C} \quad (3.13)$$

6. Fixed Input Size

$$I \in R^{224 \times 224 \times 3} \quad (3.14)$$

7. Batch Representation

$$\mathbf{X} \in R^{N \times H \times W \times C} \quad (3.15)$$

8. Example Batch Size

$$\mathbf{X} \in R^{32 \times 224 \times 224 \times 3} \quad (3.16)$$

Data Preparation

The dataset is divided into training, validation, and testing sets. Batch processing is employed to efficiently handle large datasets during training.

3.3.2 Feature Extraction

InceptionV3 Model

Feature extraction is performed using the pre-trained InceptionV3 model with `include_top=False`. This removes the classification layer and utilizes only convolutional layers for feature learning.

Feature Representation

Each image is transformed into a high-dimensional feature vector:

$$z_i = \phi(x_i) \quad (3.17)$$

where $\phi(\cdot)$ represents the feature extraction function, producing 2048-dimensional feature vectors.

3.3.3 Feature Optimization

Standardization and PCA

To ensure uniform contribution of all extracted features, standardization is applied as:

$$x' = \frac{x - \mu}{\sigma} \quad (3.18)$$

where μ and σ denote the mean and standard deviation of the feature distribution, respectively.

Dimensionality reduction is then performed using Principal Component Analysis (PCA):

$$\mathbf{z}'_i = \text{PCA}(\mathbf{z}_i) \quad (3.19)$$

The PCA process is mathematically represented as follows:

1. Mean Centering

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (3.20)$$

2. Covariance Matrix

$$\mathbf{C} = \frac{1}{n} \mathbf{X}^T \mathbf{X} \quad (3.21)$$

3. Eigenvalue Decomposition

$$\mathbf{C}\mathbf{w} = \lambda\mathbf{w} \quad (3.22)$$

4. Projection onto Principal Components

$$\mathbf{Z} = \mathbf{X}\mathbf{W} \quad (3.23)$$

5. Variance Retention Criterion

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^n \lambda_i} \geq 0.90 \quad (3.24)$$

Genetic Algorithm for Feature Selection

To identify the most relevant subset of features, a Genetic Algorithm (GA) is employed as an optimization-based feature selection technique. The GA process is mathematically described as follows:

1. Chromosome Representation

$$\mathbf{X} = [x_1, x_2, x_3, \dots, x_n], \quad x_i \in \{0, 1\} \quad (3.25)$$

2. Initial Population

$$\mathbf{P} = \{\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \dots, \mathbf{X}_m\} \quad (3.26)$$

3. Fitness Evaluation

$$\text{Fitness} = f(\mathbf{X}) \quad (3.27)$$

4. Selection

$$\mathbf{P}_{\text{selected}} \subseteq \mathbf{P} \quad (3.28)$$

5. Crossover

$$\mathbf{X}_{\text{child}} = \text{Crossover}(\mathbf{X}_1, \mathbf{X}_2) \quad (3.29)$$

6. Mutation

$$\mathbf{X}'_i = \mathbf{X}_i + \text{Mutation} \quad (3.30)$$

7. Generation Update

$$\mathbf{P}_{\text{new}} = \{\mathbf{X}'_1, \mathbf{X}'_2, \dots, \mathbf{X}'_m\} \quad (3.31)$$

8. Iteration

$$t = 1, 2, \dots, T \quad (3.32)$$

The iterative optimization process continues until the maximum number of generations is reached or convergence is achieved, resulting in the selection of the optimal feature subset.

Thus, the Genetic Algorithm effectively identifies the most relevant feature subset, improving model performance and reducing computational complexity.

3.3.4 Model Training

Model training is performed using a Bidirectional Long Short-Term Memory (BiLSTM) network to classify CT brain images. The optimized feature vectors are reshaped to match the input format required by the model. BiLSTM processes data in both forward and backward directions, allowing it to capture complex feature relationships effectively.

The model consists of BiLSTM, dropout, and dense layers, with a sigmoid activation function for binary classification. It is trained using the Adam optimizer and binary cross-entropy loss function over multiple epochs, with validation data used to monitor performance.

1. **Forget Gate**

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \quad (3.33)$$

2. **Input Gate**

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \quad (3.34)$$

3. **Cell State Update**

$$\mathbf{C}_t = \mathbf{f}_t \odot \mathbf{C}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{C}}_t \quad (3.35)$$

4. **Output Gate**

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \quad (3.36)$$

5. **Hidden State**

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{C}_t) \quad (3.37)$$

6. **Binary Cross-Entropy Loss**

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (3.38)$$

The mathematical expressions describe the internal working of the BiLSTM model during training. The input, forget, and output gate equations control the flow of information by deciding what to retain, update, or discard at each step. The cell state equation maintains long-term memory, while the hidden state represents the output passed to the next layer. Finally, the binary cross-entropy loss function measures the difference between predicted and actual values, guiding the model to update its weights and improve prediction accuracy during training.

3.3.5 Model Evaluation

The performance of all models is evaluated using standard metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.39)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.40)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.41)$$

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.42)$$

3.3.6 Final Prediction

The final prediction is obtained using the best-performing model:

$$\hat{y}_i = f(z_i^*) \quad (3.43)$$

This structured methodology ensures efficient feature extraction, dimensionality reduction, and accurate classification of stroke conditions from CT images.

4 Results & Comparative Analysis

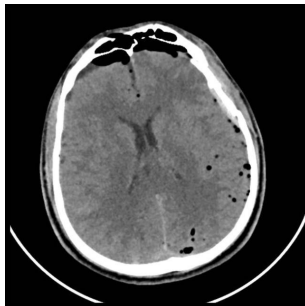
4.1 Data Collection

Dataset Source : <https://universe.roboflow.com/kaekai/brain-stroke-ct-scan-b7amc/dataset/2/download>

Table 1: Dataset Description

Attribute	Details
Dataset Name	Brain Stroke CT Scan
Total Number of Images	Normal (11843) + Stroke (5586)
Data Modality	CT Scan
Image Format	JPG
Image Resolution	640 × 640 pixels
Color Mode	Grayscale
Classes Included	Normal, Stroke

Example Images :



(a) Normal



(b) Stroke

Figure 2: **Images in the Dataset**

Data collection is a crucial step in developing an accurate and reliable stroke detection system. In this project, a dataset of brain CT scan images is collected to train and evaluate the proposed model.

The dataset consists of CT images categorized into two classes: stroke and normal. These images are obtained from publicly available medical image repositories and research datasets. CT imaging is widely used in stroke diagnosis due to its speed and effectiveness in detecting abnormalities in the brain.

4.2 Results obtained using ML Models

The performance of various Machine Learning (ML) models was evaluated for stroke detection using CT brain images. The models considered in this study include Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost. These models were trained on optimized feature vectors obtained after feature extraction and selection.

The evaluation of these models was carried out using standard performance metrics such as accuracy, precision, recall, and F1-score. The results indicate that ensemble-based models such as Random Forest and XGBoost perform better compared to traditional models like Logistic Regression and SVM. This is because ensemble methods are capable of capturing complex patterns in the data and handling feature interactions more effectively.

Classification Reports

Logistic Regression

Table 2: Performance Metrics of Logistic Regression Model

Class	Precision	Recall	F1-Score	Support
0 (Normal)	0.85	0.78	0.81	1777
1 (Stroke)	0.60	0.70	0.64	838
Accuracy	0.75			
Macro Avg	0.72	0.74	0.73	2615
Weighted Avg	0.77	0.75	0.76	2615

XGBoost

Table 3: Performance Metrics of XGBoost Model

Class	Precision	Recall	F1-Score	Support
0 (Normal)	0.84	0.83	0.83	1777
1 (Stroke)	0.65	0.65	0.65	838
Accuracy	0.77			
Macro Avg	0.74	0.74	0.74	2615
Weighted Avg	0.78	0.77	0.77	2615

Random Forest

Table 4: Performance Metrics of Random Forest

Class	Precision	Recall	F1-Score	Support
0 (Normal)	0.95	0.79	0.86	4266
1 (Stroke)	0.47	0.82	0.59	963
Accuracy	0.80			
Macro Avg	0.71	0.80	0.73	5229
Weighted Avg	0.86	0.80	0.81	5229

SVM

Table 5: Performance Metrics of SVM Model

Class	Precision	Recall	F1-Score	Support
0 (Normal)	0.90	0.89	0.90	1776
1 (Stroke)	0.78	0.79	0.78	838
Accuracy	0.86			
Macro Avg	0.84	0.84	0.84	2614
Weighted Avg	0.86	0.86	0.86	2614

Logistic Regression provides a baseline performance but is limited in capturing non-linear relationships. Support Vector Machine improves performance by using kernel functions but may struggle with large datasets. Random Forest demonstrates better accuracy due to its ability to reduce overfitting and improve generalization. XGBoost further enhances performance by using boosting techniques, leading to improved classification results.

Overall, Machine Learning models provide satisfactory performance for stroke detection; however, their ability to capture complex feature dependencies is limited compared to deep learning approaches.

4.3 Results Obtained Using DL Models

The performance of Deep Learning (DL) models was evaluated to improve the accuracy of stroke detection using CT brain images. In this study, models such as Long Short-Term Memory (LSTM) and Bidirectional Long Short-Term Memory (BiLSTM) were implemented using optimized feature vectors obtained from feature extraction and selection techniques.

The models were assessed using performance metrics including accuracy, precision, recall, and F1-score. The results show that deep learning models outperform traditional machine learning approaches due to their ability to learn complex patterns and relationships within the data.

Classification Reports

LSTM

Table 6: Performance Metrics of LSTM Model

Class	Precision	Recall	F1-Score	Support
0 (Normal)	0.83	0.73	0.78	838
1 (Stroke)	0.88	0.93	0.90	1777
Accuracy	0.86			
Macro Avg	0.85	0.83	0.84	2615
Weighted Avg	0.86	0.86	0.86	2615

BiLSTM

Table 7: Performance Metrics of Model

Class	Precision	Recall	F1-Score	Support
0 (Normal)	0.90	0.91	0.90	1777
1 (Stroke)	0.81	0.77	0.79	838
Accuracy	0.87			
Macro Avg	0.85	0.84	0.85	2615
Weighted Avg	0.87	0.87	0.87	2615

Among all the models evaluated, the BiLSTM model achieves the highest accuracy and more balanced precision and recall values for both normal and stroke classes. This indicates its effectiveness in correctly identifying stroke cases while maintaining overall reliability.

4.4 Error Analysis

Logistic Regression

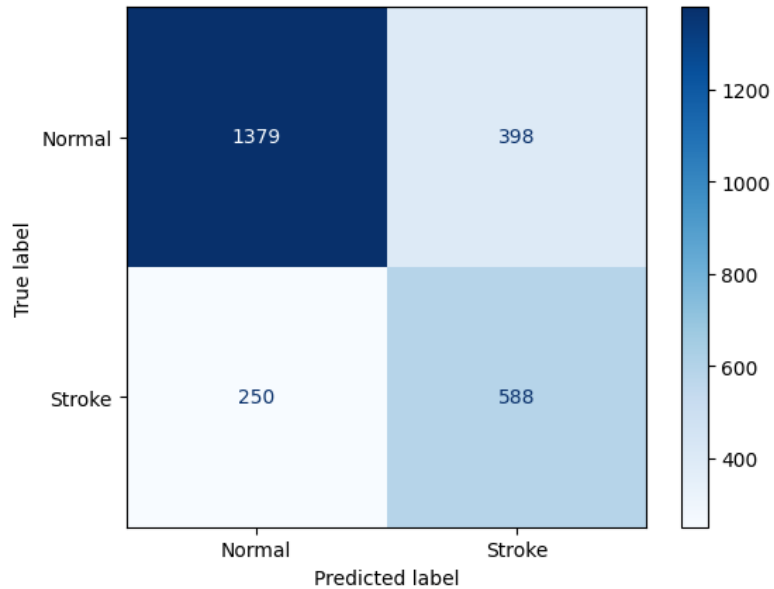


Figure 3: **Logistic Regression Confusion Matrix**

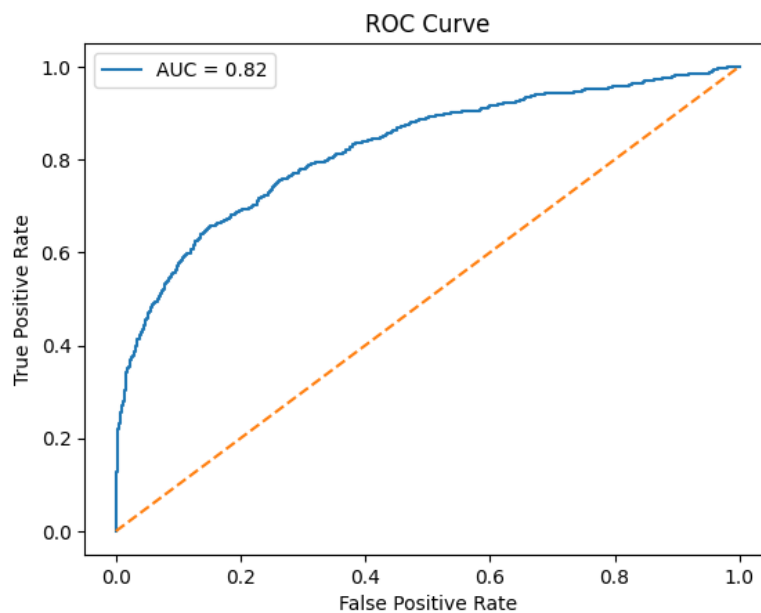


Figure 4: **Logistic Regression ROC Curve**

XGBoost

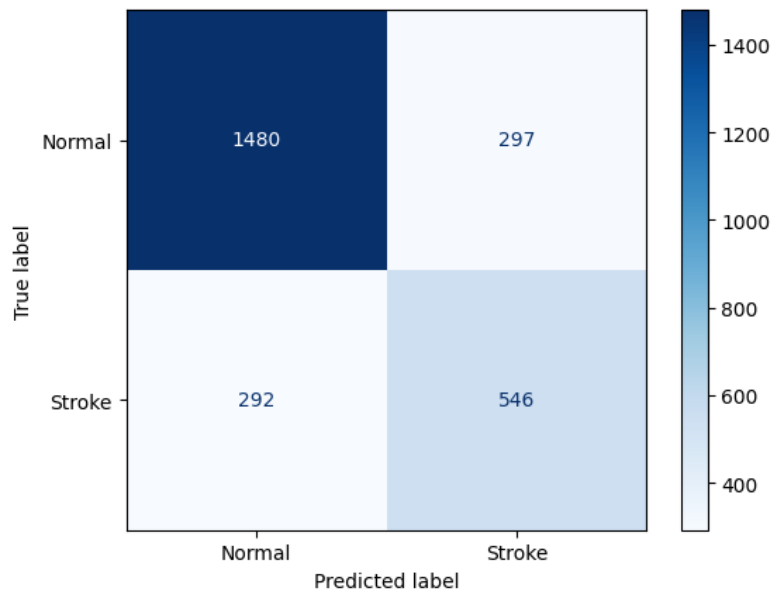


Figure 5: **XGBoost Confusion Matrix**

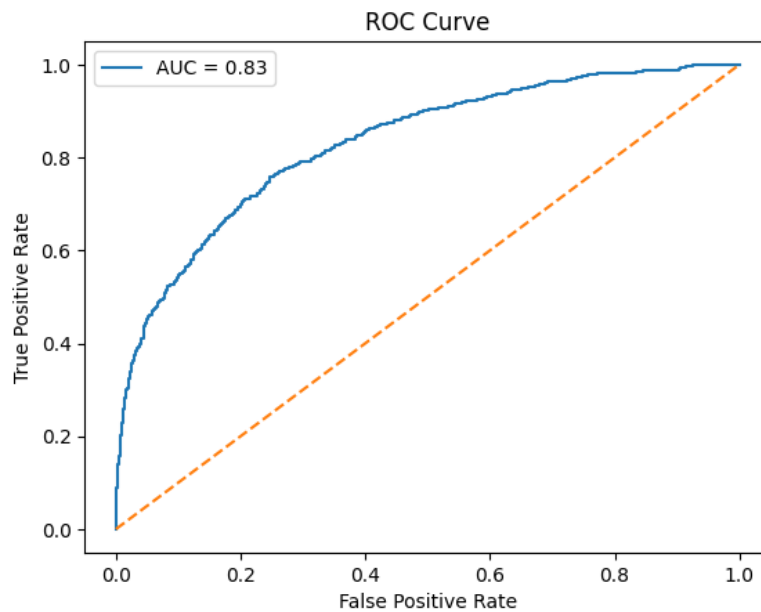


Figure 6: **XGBoost ROC Curve**

Random Forest

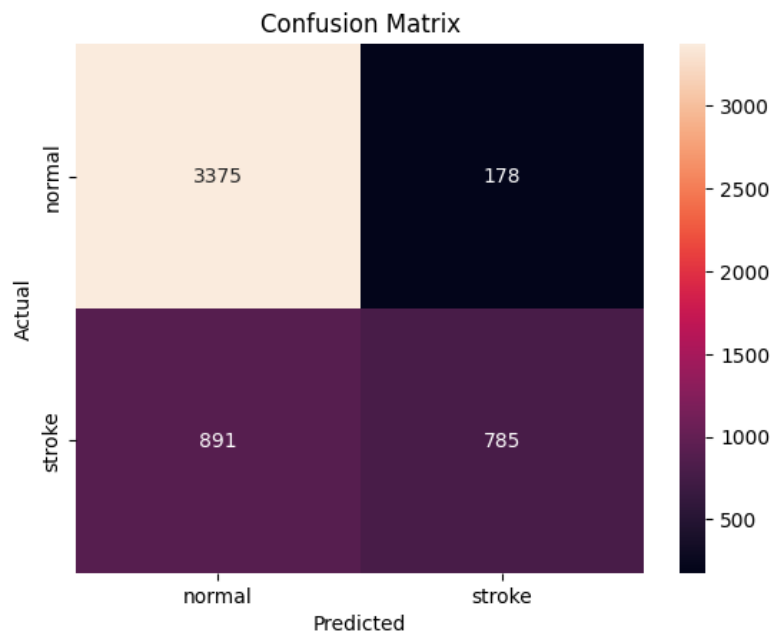


Figure 7: **Random Forest Confusion Matrix**

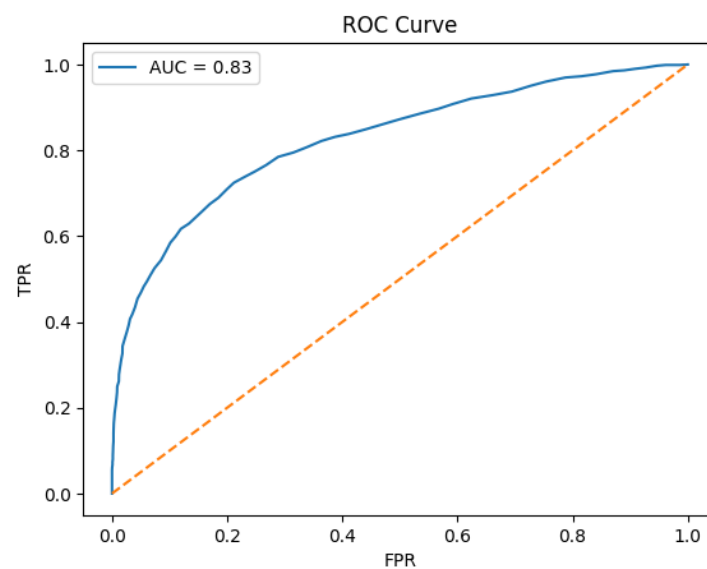


Figure 8: **Random Forest ROC Curve**

SVM

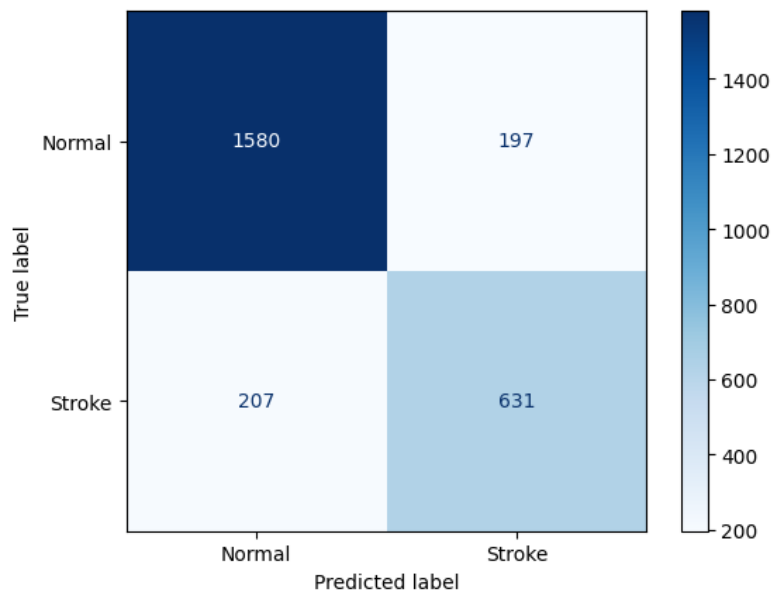


Figure 9: **SVM Confusion Matrix**

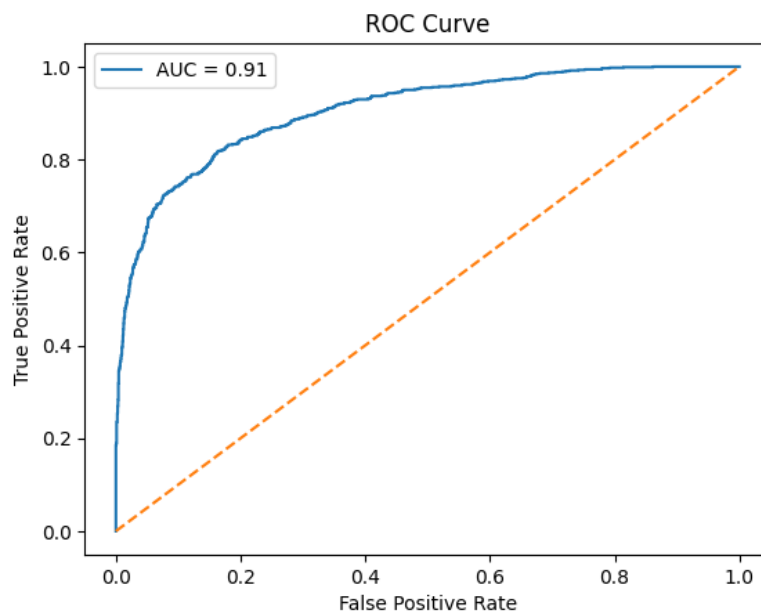


Figure 10: **SVM ROC Curve**

LSTM

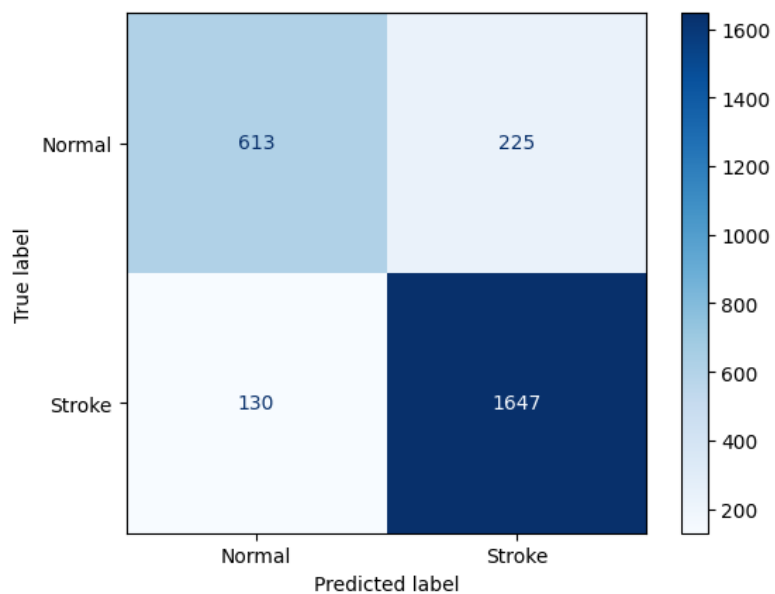


Figure 11: **LSTM Confusion Matrix**

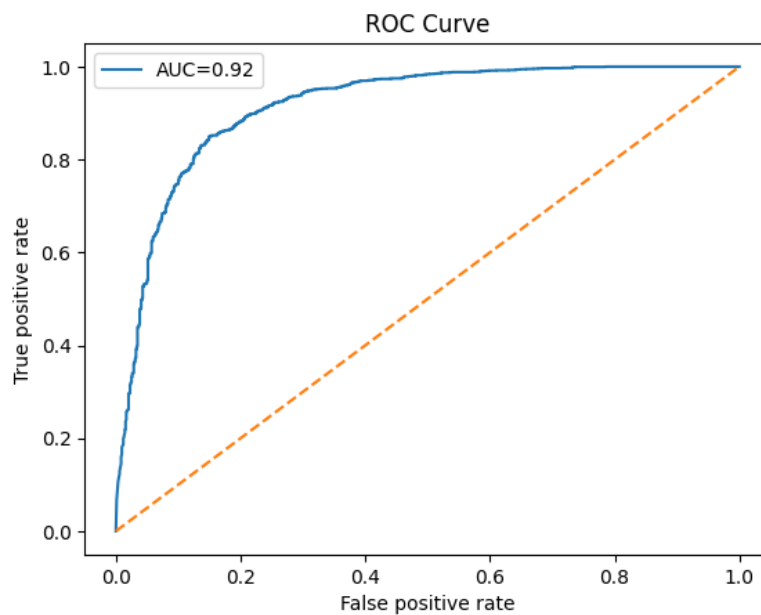


Figure 12: **LSTM ROC Curve**

BiLSTM

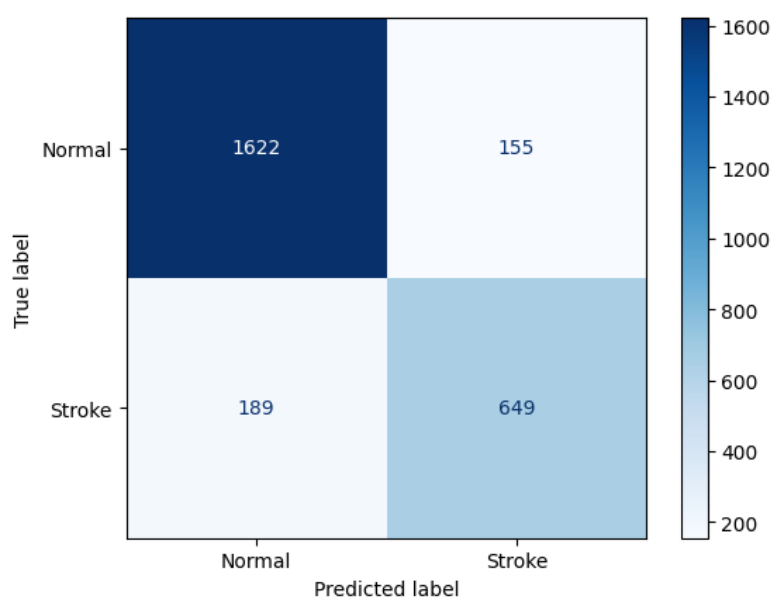


Figure 13: **BiLSTM Confusion Matrix**

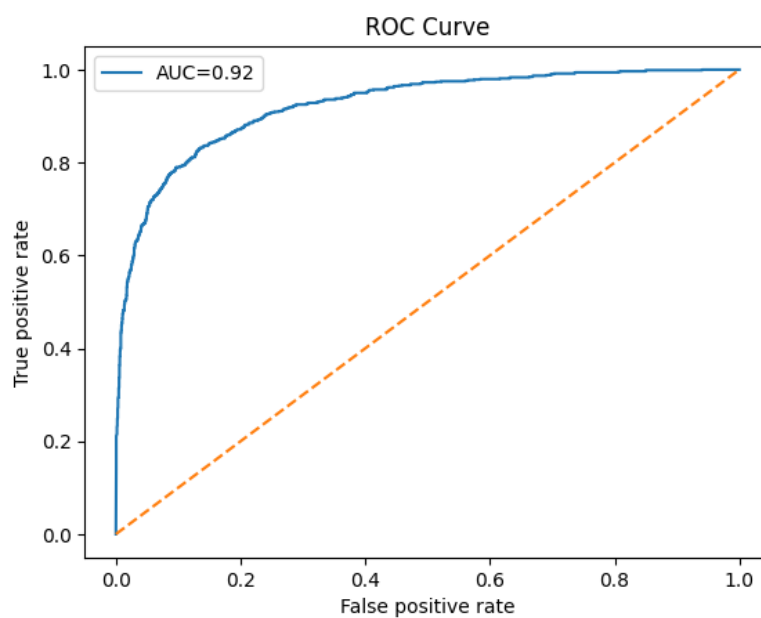


Figure 14: **BiLSTM ROC Curve**

Error analysis is carried out to understand the limitations of the proposed models and to identify the types of misclassifications occurring during prediction. The analysis is based on evaluation metrics and the confusion matrix obtained from the test results.

The model exhibits two primary types of errors: false positives and false negatives. False positives occur when normal cases are incorrectly classified as stroke, while false negatives occur when actual stroke cases are misclassified as normal. Among these, false negatives are more critical in medical applications, as missing a stroke case can delay treatment and increase the risk to patients.

The errors can be attributed to several factors. One major reason is the similarity in visual features between normal and stroke CT images, especially in early-stage stroke cases, making it difficult for the model to distinguish between them. Additionally, class imbalance in the dataset may lead the model to favor one class over the other, affecting overall performance.

Another limitation is that the model relies on features extracted from single images, without considering sequential or multi-slice information. This restricts the model's ability to capture spatial continuity and progression of stroke. Variations in image quality, noise, and differences in scanning conditions may also impact feature extraction and lead to misclassification.

Despite these limitations, the model demonstrates strong overall performance. Further improvements can be achieved by increasing dataset size, enhancing preprocessing techniques, and incorporating advanced architectures or multi-modal data.

5 Conclusion and Future Work

The proposed system presents an effective and structured approach for stroke detection using CT brain images by integrating both machine learning and deep learning techniques. Feature extraction using InceptionV3 enables the capture of rich and meaningful representations from images, while dimensionality reduction through PCA and feature selection using a Genetic Algorithm improve efficiency and reduce redundancy. A comparative analysis of multiple models demonstrates that deep learning approaches, particularly BiLSTM, achieve superior performance due to their ability to capture complex feature relationships. Overall, the system provides accurate and reliable predictions, making it a valuable tool for assisting medical professionals in early diagnosis and decision-making.

For future work, the system can be further enhanced by incorporating larger and more diverse datasets to improve generalization capability. Advanced architectures such as Vision Transformers or hybrid CNN-based models can be explored to potentially achieve higher accuracy. Real-time implementation and deployment in clinical environments can make the system more practical and impactful. Additionally, integrating multimodal data such as MRI scans and patient clinical records may further improve diagnostic performance. Further optimization through hyperparameter tuning and model fine-tuning can also contribute to improved results.

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